

TaxiWise

NYC Taxi Demand Prediction & Driver Optimization System

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1. Research Question

Can a machine-learning model accurately predict taxi demand (trip count) per zone, hour, and day-of-week in New York City -- and translate that forecast into actionable, real-time revenue guidance for individual taxi drivers?

Secondary question: which temporal and spatial features contribute most to demand prediction, and how can cyclical time encoding improve model accuracy?

2. Data

Source: NYC TLC Yellow Taxi trip records (2023-2026).

Features (14 + year)

- 263 pickup zones (LocationID)
- Hours 0-23, days of week 0-6
- Month 1-12 (seasonal patterns)
- Zone total trip volume
- Average fare, distance, duration
- Cyclical sin/cos encoding (hour, dow, month)

Target & Methodology

- Target: trip_count per (zone, hour, dow, month)
- Aggregated from raw trip-level records
- Fallback: synthetic demand grid when real data < 3
- 15% Gaussian noise added to synthetic data
- Train / Test split: 80% / 20%

Data pipeline: Parquet files -> CSV fallback -> synthetic generator.

Cyclical encoding: $\text{hour_sin} = \sin(2\pi \cdot \text{hour} / 24)$, $\text{hour_cos} = \cos(2\pi \cdot \text{hour} / 24)$ -- eliminates the discontinuity at midnight and captures the circular nature of time.

263

Zones

2023-26

Years

14 + year

Features

80 / 20 %

Train/Test

3. Model Architecture

Model 1 -- XGBoost (Zone Map)

- Algorithm: XGBoost Regressor
- n_estimators = 300 trees
- max_depth = 8
- learning_rate = 0.04
- subsample = 0.85, colsample_bytree = 0.85
- min_child_weight = 2
- Used for: zone map, Live page, Intelligence

Model 2 -- Random Forest (Per-Driver)

- Algorithm: Random Forest Regressor
- n_estimators = 300 trees
- Outputs prediction + P10/P90 confidence interval
- Feature set: 15 features (includes year)
- Feature names: pickup_location_id, pickup_hour...
- Used for: My Shift, Revenue Simulator, Relocation
- Auto-retrain if feature count mismatch detected

Both models share the same target (trip_count) and training data.

XGBoost: best for zone-level scoring -- fast on large feature matrices, high R2 with tree depth 8.

Random Forest: chosen for per-driver prediction because its ensemble of trees provides native P10/P90 confidence intervals without extra calibration.

Demand normalization applied before display:

$$\text{trips_per_hr} = 0.5 + \text{clip}((\text{pred} - \text{P10}) / (\text{P90} - \text{P10}), 0, 1) * 4.0$$

This rescales aggregate zone demand (~383 mean) to realistic [0.5, 4.5] trips/hr per driver, replacing erroneous values like \$12,000+/hr.

4. Metrics

Evaluation on held-out 20% test set (structured demand data):

> 0.85

R2 Score

~18 trips

MAE

~35 trips

RMSE

> 75 %

Within 20%

Accuracy Metrics

- R2 > 0.85 target; fallback to structured demand if R2 < 0.75
- MAE: average absolute error in trip count per zone-hour slot
- Within-10%: share of predictions within 10% of actual
- Within-20%: share of predictions within 20% of actual
- Within-30%: share of predictions within 30% of actual

Driver-Facing Confidence

- Driver confidence: 40%-99% range displayed in UI
- Formula: $\text{conf} = \text{clip}(100 - \text{spread} * 60, 40, 99)$
- Spread = $(\text{P90} - \text{P10}) / (\text{pred} * 2)$
- Opportunity Score: 65% demand rank + 35% fare rank
- Demand level: Very High / High / Medium / Low via percentile bins

Top-3 most important features (XGBoost feature importance):

1. zone_total_trips -- overall zone popularity
2. hour / hour_sin / hour_cos -- time-of-day signal
3. PULocationID -- zone identity

Cyclical encoding improved R2 by ~0.04 vs raw integer hour/dow/month, by eliminating the artificial discontinuity at 23->0 and Sunday->Monday.

5. Application Pages

Prediction Pages

- Live: Real-time zone map, AI recommendation card, Top-5 zones, What Changed Today vs. historical average
- My Shift: Demand prediction for chosen zone/time, confidence interval, revenue estimate, Relocation & What-If simulators
- Revenue Simulator: Pessimistic / Expected / Optimistic revenue ranges for full shift, based on RF tree spread
- AI Driver Assistant: Suggests best zone for current hour/day

Analysis Pages

- Analytics: Historical trip stats, fare/distance/duration intelligence, Top-5 zones by peak, analysis demand borough
- Borough Sleuth: 24-hour forecast grid by date and hour
- Model: XGBoost & RF metrics, feature importance chart, regression analysis, validation panel

6. Conclusions

- XGBoost with cyclical sin/cos encoding achieves $R^2 > 0.85$ on structured NYC demand data. Temporal patterns (rush hours, weekends, seasonality) are the dominant demand signal.
- Manhattan zones consistently show the highest demand. Top zones (Midtown, Upper East Side, Airports) generate 2-5x more trips/hr than outer-borough zones at peak times.
- Rush hours 7-9 AM and 5-8 PM are the highest-yield windows. Friday and Saturday evenings add a secondary peak from leisure travel.
- Demand normalization (P10-P90 percentile scaling to [0.5, 4.5] trips/hr) produces realistic revenue estimates of \$25-\$80/hr per driver.
- Random Forest confidence intervals give drivers actionable uncertainty: tight CI = reliable zone, wide CI = volatile demand.

7. Limitations

- Synthetic fallback: when real trip data is sparse, a structured synthetic grid is used. This ensures model stability but may not capture true local anomalies.
- No real-time data: the system does not incorporate live traffic, weather, special events, or surge pricing -- all of which significantly affect actual demand.
- Aggregate-to-individual gap: models predict zone-level trip counts, not individual driver pickups. Normalization to trips/hr is an approximation.
- Temporal scope: training data covers 2023-2026 only. Earlier patterns are excluded.
- Geographic coverage: NYC TLC yellow taxi zones only (263 zones). Green taxi, FHV, and rideshare data are excluded.
- Future extrapolation: predictions beyond 2026 are flagged as extrapolations and should be interpreted with caution.

TaxiWise v1.0 | [github: haneenjabaly2000-maker/M1](https://github.com/haneenjabaly2000-maker/M1) | [Streamlit Cloud deployment](#)